

Short-Term Momentum and Reversal Analysis on Nifty 50 Constituents

Statistical Market Signal Analysis and Forward Return Evaluation

Data period: 2025-05-26 to 2026-05-22 | Universe: 50 Nifty 50 stocks

Metric	Value
Stocks analyzed	50
Daily stock-price rows	12,294
Benchmark rows	246
Bullish streak signals	239
Bearish streak signals	199
Average next 5D return after bullish signals	0.10%
Average next 5D return after bearish signals	-0.13%

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1. Abstract

This project presents an exploratory quantitative finance study on short-term momentum and reversal behavior in Nifty 50 stocks. The system identifies strong three-day bullish and bearish price movements, then evaluates what happened over the next three and five trading days. The project combines market data collection, return calculation, signal detection, statistical aggregation, benchmark comparison, visualization, and Excel report generation in a Streamlit dashboard.

The analysis uses one year of daily OHLCV data for 50 Nifty 50 constituents and compares forward performance with the Nifty 50 benchmark. The final result is not a prediction model or trading recommendation; it is a statistical research framework for understanding whether recent short-term moves tend to continue or reverse.

2. Introduction and Problem Statement

Financial markets produce large volumes of price data every day. Manual review of this data can be inconsistent, slow, and affected by emotional bias. Quantitative analysis uses mathematical and statistical rules to study price behavior in a structured way.

The central question in this project is: after a Nifty 50 stock makes a strong move over three consecutive trading days, does the stock continue in the same direction, reverse, or behave randomly over the next few days?

3. Objectives and Scope

- Collect and process one year of daily Nifty 50 OHLCV data.
- Calculate daily returns, three-day returns, and forward three-day and five-day returns.
- Detect bullish and bearish streak signals using a threshold-based rule.
- Measure stock-wise continuation, reversal, bounce, volatility, and benchmark alpha.

- Create a Streamlit dashboard and downloadable Excel report for practical use.
- Document the methodology clearly enough for learning and presentation.

4. Tools and Libraries Used

Tool / Library	Use in Project
Python	Main programming language used for data processing and app development.
Pandas	Data cleaning, grouping, return calculation, summary tables, CSV handling.
NumPy	Numerical calculations and conditional signal logic.
yfinance	Download historical stock and benchmark data from Yahoo Finance.
Requests	Fetch the official Nifty 50 constituent list from NSE archives.
Streamlit	Interactive dashboard interface for running the analysis.
Plotly	Histograms, threshold charts, and heatmap visualizations.
OpenPyXL	Formatted Excel report generation.
Excel / CSV	Portable tabular outputs for reporting and presentation.

5. Statistical and Mathematical Framework

This project does not use machine learning models. It uses statistical and quantitative finance methods. The main methods are return calculation, threshold-based signal detection, forward return analysis, win-rate measurement, volatility measurement, benchmark alpha, and threshold sensitivity testing.

5.1 Return Calculation

```
Daily Return_t = ((Close_t - Close_(t-1)) / Close_(t-1)) * 100
3-Day Return_t = ((Close_t / Close_(t-3)) - 1) * 100
```

5.2 Signal Detection

```
Bullish Signal: 3-Day Return >= +5%
Bearish Signal: 3-Day Return <= -5%
Optional filter: all three daily returns must move in the same direction.
```

5.3 Forward Return Analysis

```
Forward 3D Return_t = ((Close_(t+3) / Close_t) - 1) * 100
Forward 5D Return_t = ((Close_(t+5) / Close_t) - 1) * 100
```

5.4 Benchmark Alpha

5D Alpha vs Nifty = Stock Forward 5D Return - Nifty Forward 5D Return

6. System Workflow and Dashboard Design

1.

Load the Nifty 50 stock universe and one year of historical price data.
2.

Clean and align stock-level OHLCV data with benchmark data.
3.

Compute return features and detect bullish or bearish streak signals.
4.

Calculate forward returns and classify continuation or reversal behavior.
5.

Aggregate stock-wise statistics and create ranking tables.
6.

Display the results in Streamlit and export the final report to Excel.

Top Bearish Continuation											
Stock	Bullish Signal Count	Avg Next 3D After Bullish %	Avg Next 5D After Bullish %	Median Next 5D After Bullish %	Std Dev Next 5D After Bullish %	Best Next 5D After Bullish %	Worst Next 5D After Bullish %	Bullish Continuation Rate %	Avg 5D Alpha After Bullish %	Bears	
BAJAJFINSVS	2	-1.04	-1.65	-1.65	5.92	2.54	-5.84	50	-0.18		
HINDUNILVR.NS	1	-1.43	-0.58	-0.58	None	-0.58	-0.58	0	1.62		
INDIGO.NS	5	0.13	0.85	1.07	4.01	6.39	-3.82	60	-0.64		
IT.NS	2	3.61	4.7	4.7	1.95	6.08	3.32	100	2.35		
HDFCBANK.NS	2	0.42	0.27	0.27	3.96	3.07	-2.53	50	-2.05		
ITC.NS	1	-2.12	-3.82	-3.82	None	-3.82	-3.82	0	-2.52		
BEL.NS	2	0.93	0.5	0.5	4.06	3.37	-2.37	50	-1.37		
SBIN.NS	1	0.79	2.98	2.98	None	2.98	2.98	100	-0.82		
RELIANCE.NS	3	-0.21	0.07	0.88	1.84	1.37	-2.03	66.67	0.61		
HCLTECH.NS	3	0.5	1.54	0.67	2.44	4.3	-0.35	66.67	0.15		

Figure 1: Dashboard overview with key project metrics and automatic conclusion.

7. Data Collection and Processing

The final analysis used 50 stocks, 12,294 stock-price rows, and 246 benchmark rows. The dataset covers the period from 2025-05-26 to 2026-05-22.

Dataset	Rows	Main Fields
Price Data Used	12,294	Date, Stock, Open, High, Low, Close, Volume
Benchmark Data Used	246	Date, Stock, Open, High, Low, Close, Volume
Bullish Signals	239	Signal date, return features, forward returns, result, alpha
Bearish Signals	199	Signal date, return features, forward returns, result, alpha

8. Experimental Results

The system detected 239 bullish streak signals and 199 bearish streak signals. After bullish streaks, the average next five-day return was 0.10% with a 48.5% continuation rate. After bearish streaks, the average next five-day return was -0.13% with a 47.7% bounce rate.

The average five-day alpha versus Nifty was -0.04% after bullish signals and 0.08% after bearish signals. This suggests that, at the aggregate level, the observed effects were modest and should be interpreted cautiously.

8.1 Top Bullish Continuation Stocks

Stock	Signals	Avg Next 5D %	Continuation %	Alpha %
LT.NS	2	4.70	100.00	2.35
TMPV.NS	4	2.78	100.00	2.30
WIPRO.NS	2	2.67	100.00	-1.63
BAJFINANCE.NS	9	2.49	66.67	1.38
ASIANPAINT.NS	8	2.38	62.50	1.85

8.2 Top Bullish Reversal Stocks

Stock	Signals	Avg Next 5D %	Continuation %	Alpha %
COALINDIA.NS	4	-3.48	0.00	-2.60
TECHM.NS	4	-3.22	25.00	-3.26
CIPLA.NS	6	-2.55	33.33	-2.84
INFY.NS	3	-2.33	0.00	-2.24
M&M.NS	6	-2.31	33.33	-1.68

8.3 Top Bearish Bounce Stocks

Stock	Signals	Avg Next 5D %	Bounce %	Alpha %
HINDALCO.NS	6	3.23	100.00	3.54
TATASTEEL.NS	3	3.15	66.67	4.31
M&M.NS	7	2.12	71.43	1.64
GRASIM.NS	2	1.90	100.00	3.24
TRENT.NS	6	1.51	83.33	1.04

8.4 Top Bearish Continuation Stocks

Stock	Signals	Avg Next 5D %	Bounce %	Alpha %
BAJAJFINSV.NS	4	-3.02	0.00	-1.93
HINDUNILVR.NS	2	-2.76	0.00	-1.55
INDIGO.NS	10	-2.67	40.00	-2.27
LT.NS	3	-1.98	0.00	-0.28
HDFCBANK.NS	4	-1.85	25.00	-0.99

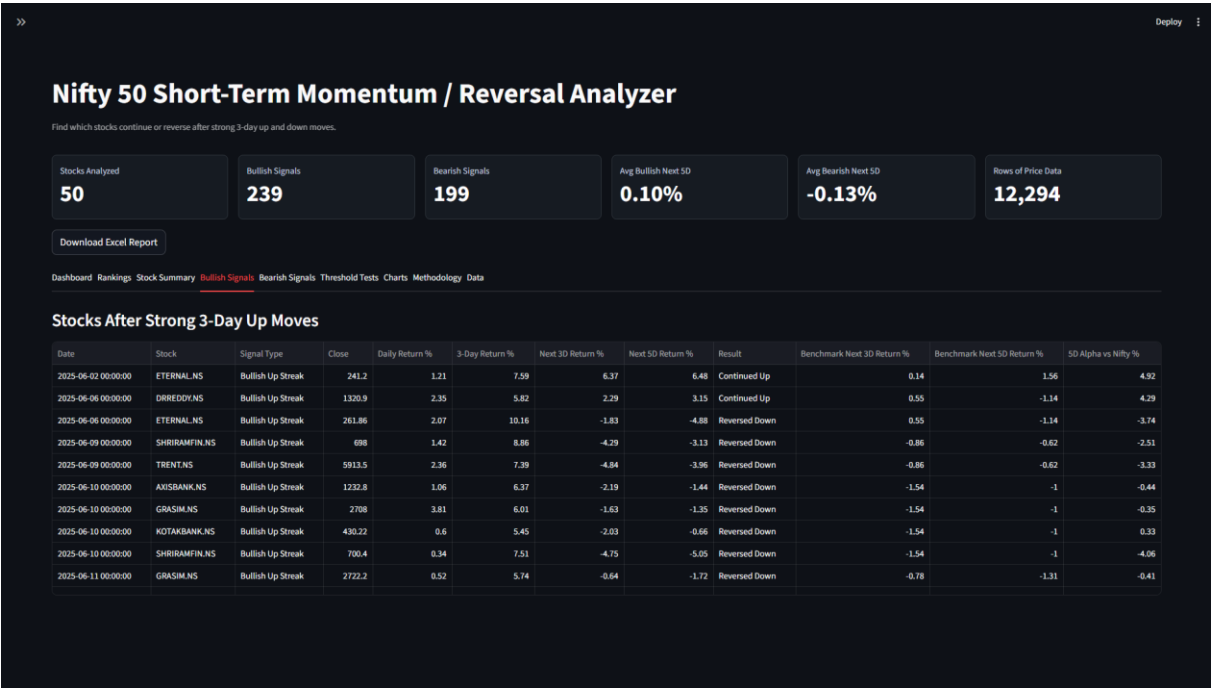


Figure 2: Ranking tables generated in the dashboard.

9. Threshold Sensitivity Testing

Threshold sensitivity testing checks whether conclusions change when the signal threshold is changed. The dashboard tested 3%, 4%, 5%, and 6% three-day return thresholds.

Threshold %	Bullish Signals	Avg Bullish 5D %	Bullish Cont. %	Bearish Signals	Avg Bearish 5D %	Bearish Bounce %
3.00	632	-0.02	47.94	577	-0.06	48.35
4.00	397	0.03	47.36	343	-0.18	46.36
5.00	239	0.10	48.54	199	-0.13	47.74
6.00	146	0.23	52.05	131	0.07	51.91

The 6% threshold produced fewer signals but stronger average forward results. This is expected because a stricter signal definition filters out weaker price moves. However, fewer signals also means a smaller sample size, so the results should not be over-interpreted.

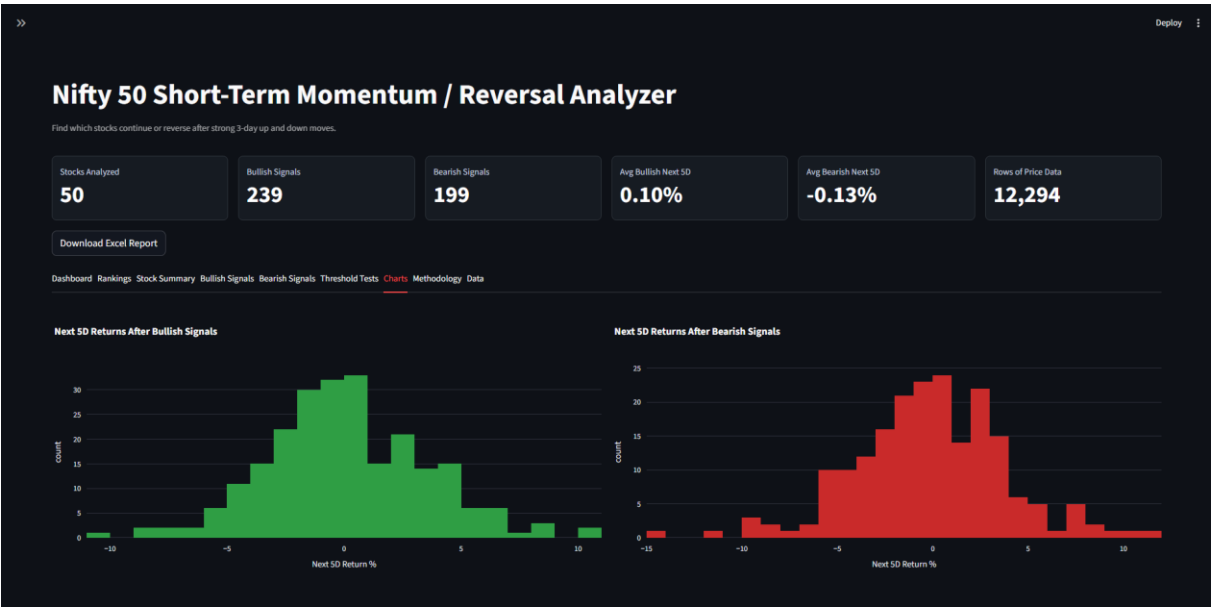


Figure 3: Threshold sensitivity table in the dashboard.

10. Visual Analysis

The dashboard includes histograms to show the distribution of future returns after bullish and bearish streaks. These charts are useful because average return alone can hide wide variation and outliers.

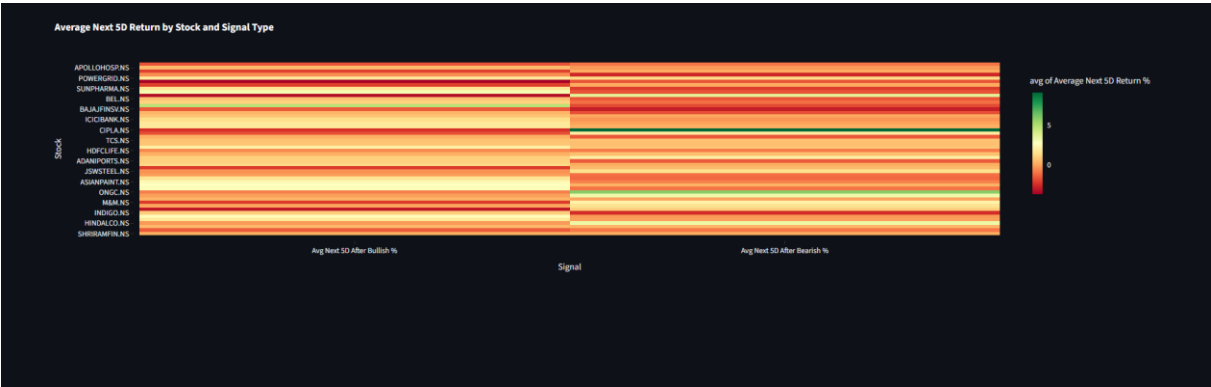


Figure 4: Distribution of next five-day returns after bullish and bearish signals.

The heatmap compares average next five-day behavior across stocks and signal types. It helps identify which stocks are more momentum-oriented and which show reversal tendencies.

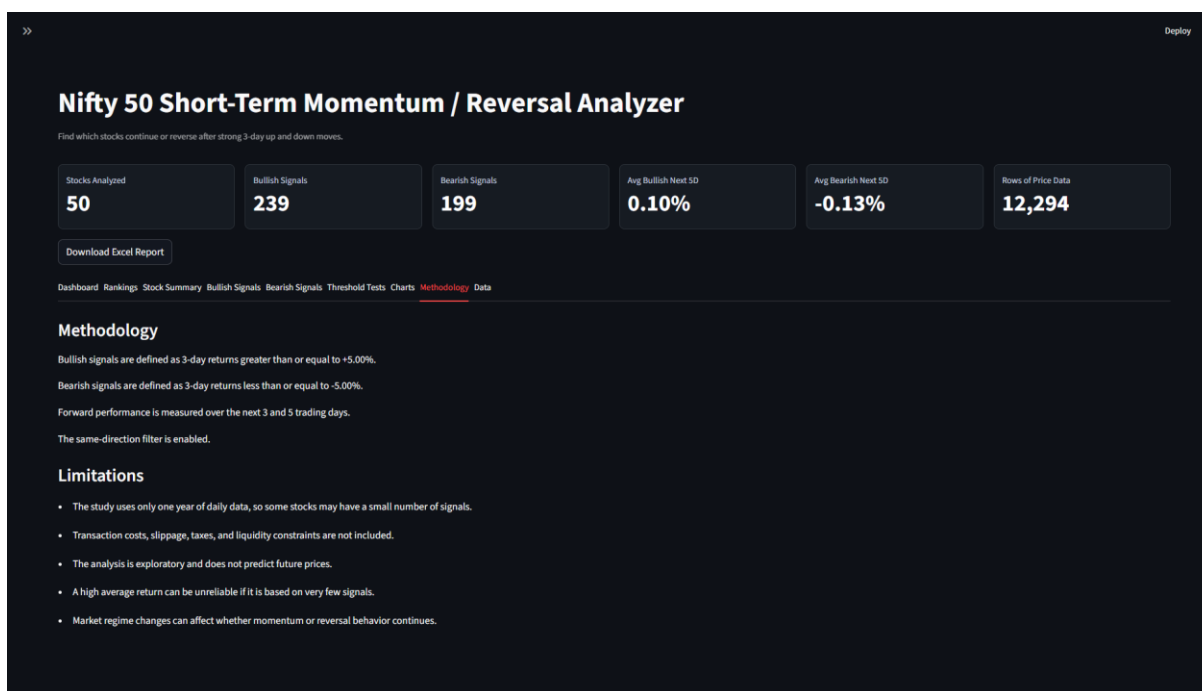


Figure 5: Heatmap of average forward return by stock and signal type.

11. Interpretation and Findings

- Bullish signals had a slightly positive average next five-day return of 0.10%, but the continuation rate was below 50%. This indicates weak aggregate momentum.
- Bearish signals had a slightly negative average next five-day return of -0.13%, with a bounce rate below 50%. This suggests no strong aggregate mean-reversion effect at the 5% threshold.
- Some stocks showed clear stock-specific behavior. For example, LT.NS ranked highest in bullish continuation, while HINDALCO.NS ranked highest in bearish bounce.
- The effect becomes more positive at the 6% threshold, but the sample size becomes smaller.
- The study is best treated as an exploratory statistical scan, not as a standalone trading strategy.

12. Limitations

- Only one year of daily data is used, so the sample may not represent all market regimes.
- Some stocks have fewer signals, making their rankings less statistically reliable.
- Transaction costs, slippage, brokerage, taxes, and liquidity are not included.
- The analysis does not include risk-adjusted portfolio construction or position sizing.
- The project does not forecast prices; it only studies what happened historically after defined patterns.
- Nifty 50 constituents and ticker availability may change over time.

13. Future Scope

- Extend the study to three or five years of data to improve statistical reliability.
- Compare different holding periods such as 1 day, 10 days, and 20 days.

- Add sector-wise analysis to detect whether some industries are more momentum-driven.
- Add risk-adjusted metrics such as Sharpe ratio and maximum drawdown.
- Include transaction cost simulation before converting the study into any strategy idea.
- Add hypothesis testing to check whether average forward returns are statistically significant.

14. Conclusion

This project successfully builds a complete quantitative market analytics workflow for Nifty 50 stocks. It collects data, detects short-term bullish and bearish streaks, measures future returns, compares results with a benchmark, ranks stocks, visualizes distributions, and exports structured reports.

The results show that aggregate continuation and reversal effects are modest over the selected one-year period. However, stock-wise ranking reveals that individual stocks can show stronger tendencies. The project is therefore valuable as a learning-oriented quantitative analysis system and as a foundation for deeper research.

15. Appendix

15.1 Dashboard Methodology Screen

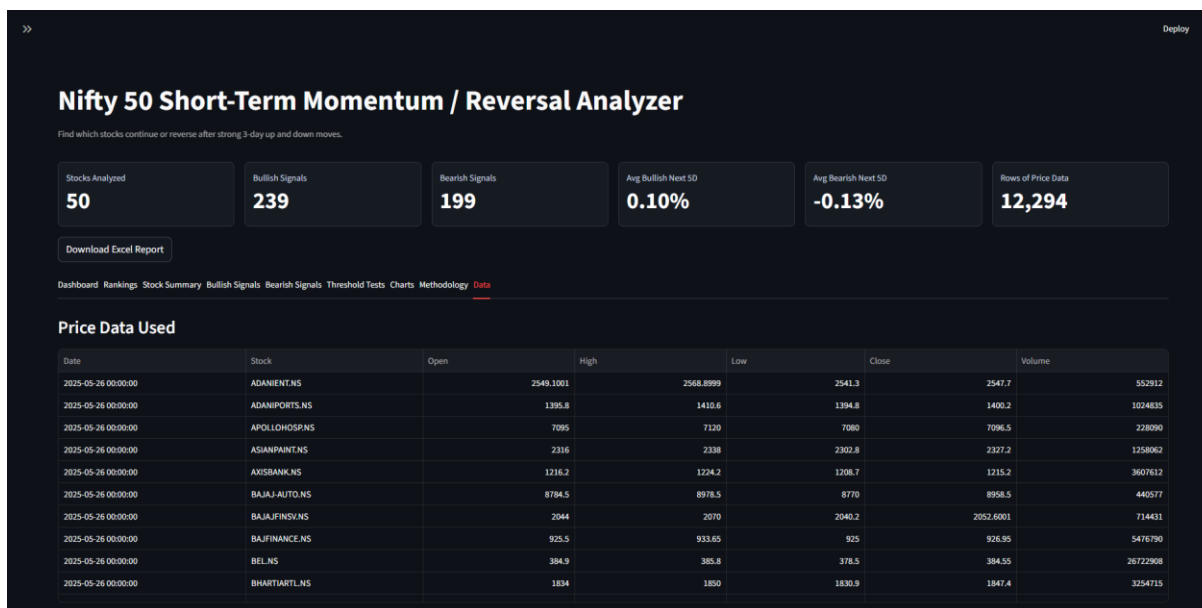


Figure 6: Dashboard methodology and limitations screen.

15.2 Sample Signal Tables

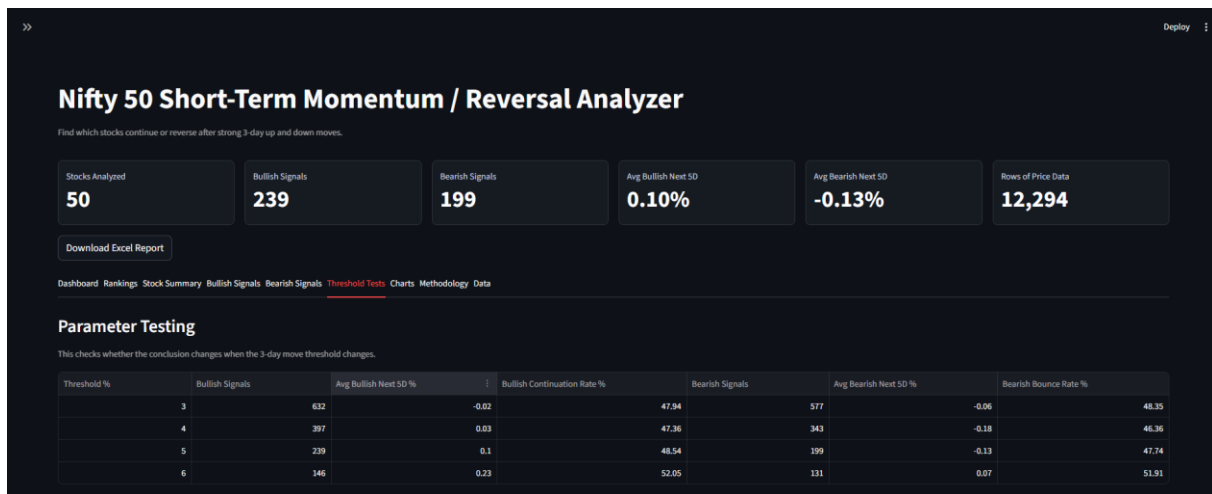


Figure 7: Sample bullish signal table.

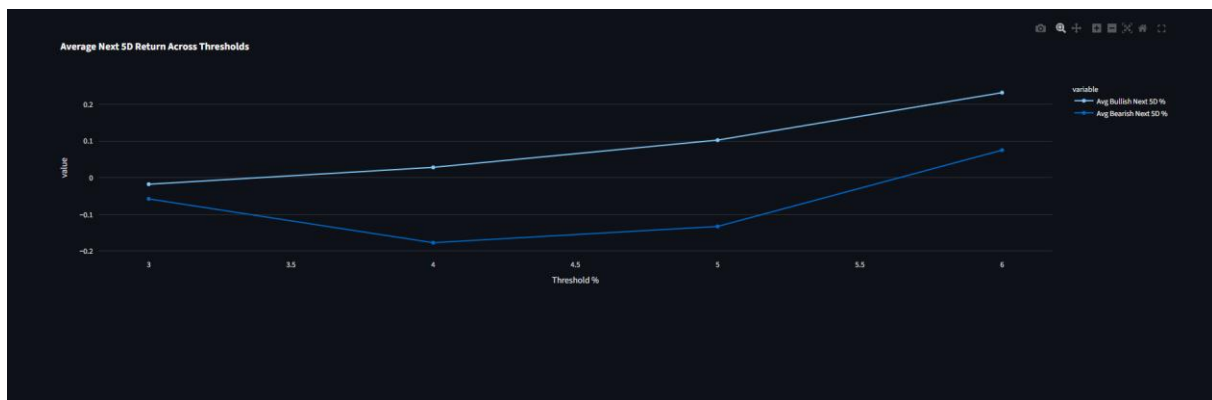


Figure 8: Sample bearish signal table.

15.3 Learning Notes

- Daily return measures one-day percentage price movement.
- Three-day return identifies short-term strength or weakness.
- Forward return checks what happened after the signal date.
- Continuation means price moved further in the same direction.
- Reversal or bounce means price moved opposite to the prior streak.
- Standard deviation measures variation in outcomes.
- Alpha compares stock performance against the Nifty 50 benchmark.